

Task 1: Create a New EBS Volume

In this task, you will create and attach an Amazon EBS volume to a new Amazon EC2 instance.

4. In the AWS Management Console, in the search box next to **Services**, search for and select **EC2**.
5. In the left navigation pane, choose **Instances**.

An Amazon EC2 instance named **Lab** has already been launched for your lab.

6. Note the **Availability Zone** of the instance. It will look similar to *us-east-1a*.
7. In the left navigation pane, choose **Volumes**.

You will see an existing volume that is being used by the Amazon EC2 instance. This volume has a size of 8 GiB, which makes it easy to distinguish from the volume you will create next, which will be 1 GiB in size.

8. Choose **Create volume** then configure:
 - **Volume Type:** *General Purpose SSD (gp2)*
 - **Size (GiB):** 1. **NOTE:** You may be restricted from creating large volumes.
 - **Availability Zone:** Select the same availability zone as your EC2 instance.
 - Choose **Add tag**
 - In the Tag Editor, enter:
 - **Key:** Name
 - **Value:** My Volume
9. Choose **Create Volume**

Your new volume will appear in the list, and will move from the *Creating* state to the *Available* state. You may need to choose **refresh** to see your new volume.

Task 2: Attach the Volume to an Instance

In this task you will attach the new EBS volume to the Amazon EC2 instance.

10. Select **My Volume**.

11. In the **Actions** menu, choose **Attach volume**.

12. Choose the **Instance** field, then select the **Lab** instance.

Note that the **Device** name is set to `/dev/sdf`. Notice also the message displayed that "Newer Linux kernels may rename your devices to `/dev/xvdf` through `/dev/xvdp` internally, even when the device name entered here (and shown in the details) is `/dev/sdf` through `/dev/sdp`."

13. Choose **Attach volume**

The volume state is now *In-use*.

Task 3: Connect to Your Amazon EC2 Instance

In this task, you will connect to the EC2 instance using EC2 Instance Connect which provides access to a terminal in the browser.

14. In the AWS Management Console, in the search box next to **Services**, search for and select **EC2**.

15. Choose **Instances**.

16. Select the **Lab** instance, and then choose **Connect**.

17. On the **EC2 Instance Connect** tab, choose **Connect**.

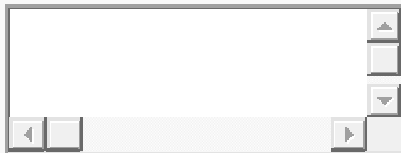
An EC2 Instance Connect terminal session opens and displays a `$` prompt.

Task 4: Create and Configure Your File System

In this task, you will add the new volume to a Linux instance as an ext3 file system under the `/mnt/data-store` mount point.

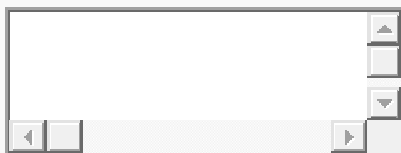
18. View the storage available on your instance:

Run the following command:



```
df -h
```

You should see output similar to:

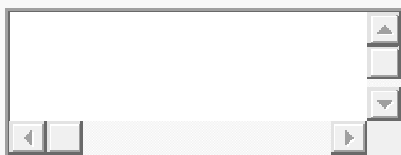


Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	4.0M	0	4.0M	0%	/dev
tmpfs	475M	0	475M	0%	/dev/shm
tmpfs	190M	2.8M	188M	2%	/run
/dev/xvda1	8.0G	1.6G	6.5G	20%	/
tmpfs	475M	0	475M	0%	/tmp
tmpfs	95M	0	95M	0%	/run/user/1000

The output shows that the original 8GB **/dev/xvda1** disk volume mounted at `/` which indicates that it is the root volume. It hosts the Linux operating system of the EC2 instance.

The 1GB other volume that you attached to the Lab instance is not listed, because you have not yet created a file system on it or mounted the disk. Those actions are necessary so that Linux operating system can make use of the new storage space. You will take those actions next.

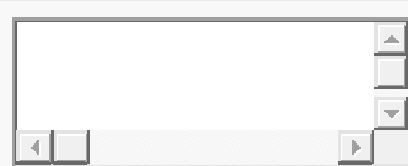
19. Create an ext3 file system on the new volume:



```
sudo mkfs -t ext3 /dev/sdf
```

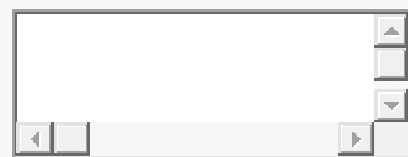
The output should indicate that a new file system was created on the attached volume.

20. Create a directory for mounting the new storage volume:



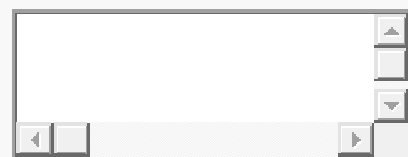
```
sudo mkdir /mnt/data-store
```

21. Mount the new volume:



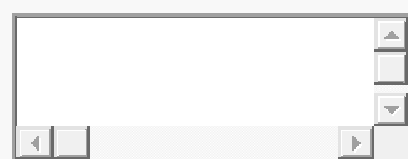
```
sudo mount /dev/sdf /mnt/data-store
```

To configure the Linux instance to mount this volume whenever the instance is started, you will need to add a line to */etc/fstab*. Run the command below to accomplish that:



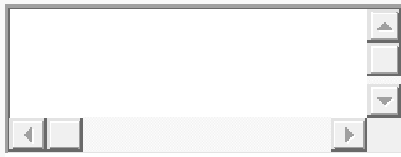
```
echo "/dev/sdf /mnt/data-store ext3 defaults,noatime 1 2" | sudo tee -a /etc/fstab
```

22. View the configuration file to see the setting on the last line:



```
cat /etc/fstab
```

23. View the available storage again:



```
df -h
```

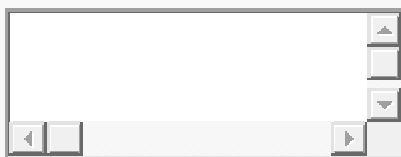
The output will look similar to what is shown below.



Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	484M	0	484M	0%	/dev
tmpfs	492M	0	492M	0%	/dev/shm
tmpfs	492M	460K	491M	1%	/run
tmpfs	492M	0	492M	0%	/sys/fs/cgroup
/dev/xvda1	8.0G	1.5G	6.6G	19%	/
tmpfs	99M	0	99M	0%	/run/user/0
tmpfs	99M	0	99M	0%	/run/user/1000
/dev/xvdf	976M	1.3M	924M	1%	/mnt/data-store

Notice the last line. The output now lists */dev/xvdf* which is the new mounted volume.

24. On your mounted volume, create a file and add some text to it.



```
sudo sh -c "echo some text has been written > /mnt/data-store/file.txt"
```

25. Verify that the text has been written to your volume.



```
cat /mnt/data-store/file.txt
```

Leave the EC2 Instance Connect session running. You will return to it later in this lab.

Task 5: Create an Amazon EBS Snapshot

In this task, you will create a snapshot of your EBS volume.

You can create any number of point-in-time, consistent snapshots from Amazon EBS volumes at any time. Amazon EBS snapshots are stored in Amazon S3 with high durability. New Amazon EBS volumes can be created out of snapshots for cloning or restoring backups. Amazon EBS snapshots can also be easily shared among AWS users or copied over AWS regions.

26. In the **EC2 Console**, choose **Volumes** and select **My Volume**.

27. In the **Actions** menu, select **Create snapshot**.

28. Choose **Add tag** then configure:

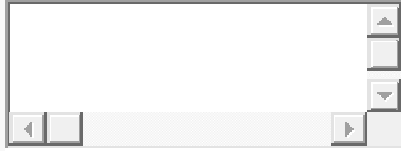
- **Key:** Name
- **Value:** My Snapshot
- Choose **Create snapshot**

29. In the left navigation pane, choose **Snapshots**.

Your snapshot is displayed. The status will first have a state of *Pending*, which means that the snapshot is being created. It will then change to a state of *Completed*.

Note: Only used storage blocks are copied to snapshots, so empty blocks do not occupy any snapshot storage space.

30. In your EC2 Instance Connect session, delete the file that you created on your volume.



```
sudo rm /mnt/data-store/file.txt
```

31. Verify that the file has been deleted.



```
ls /mnt/data-store/
```

Your file has been deleted.

Task 6: Restore the Amazon EBS Snapshot

If you ever wish to retrieve data stored in a snapshot, you can **Restore** the snapshot to a new EBS volume.

Create a Volume Using Your Snapshot

32. In the **EC2 console**, select **My Snapshot**.

33. In the **Actions** menu, select **Create volume from snapshot**.

34. For **Availability Zone**, select the same availability zone that you used earlier.

35. Choose **Add tag** then configure:

- **Key:** Name
- **Value:** Restored volume
- Choose **Create volume**

Note: When restoring a snapshot to a new volume, you can also modify the configuration, such as changing the volume type, size or Availability Zone.

Attach the Restored Volume to Your EC2 Instance

36. In the left navigation pane, choose **Volumes**.

37. Select **Restored Volume**.

38. In the **Actions** menu, select **Attach volume**.

39. Choose the **Instance** field, then select the **Lab** instance that appears.

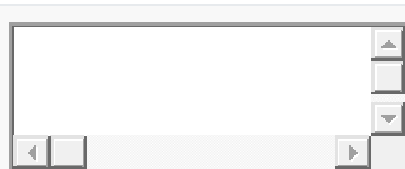
Note that the **Device** field is set to `/dev/sdg`. You will use this device identifier in a later task.

40. Choose **Attach volume**

The volume state is now *in-use*.

Mount the Restored Volume

41. Create a directory for mounting the new storage volume:



```
sudo mkdir /mnt/data-store2
```

42. Mount the new volume:



```
sudo mount /dev/sdg /mnt/data-store2
```

43. Verify that volume you mounted has the file that you created earlier.



```
ls /mnt/data-store2/
```

You should see file.txt.